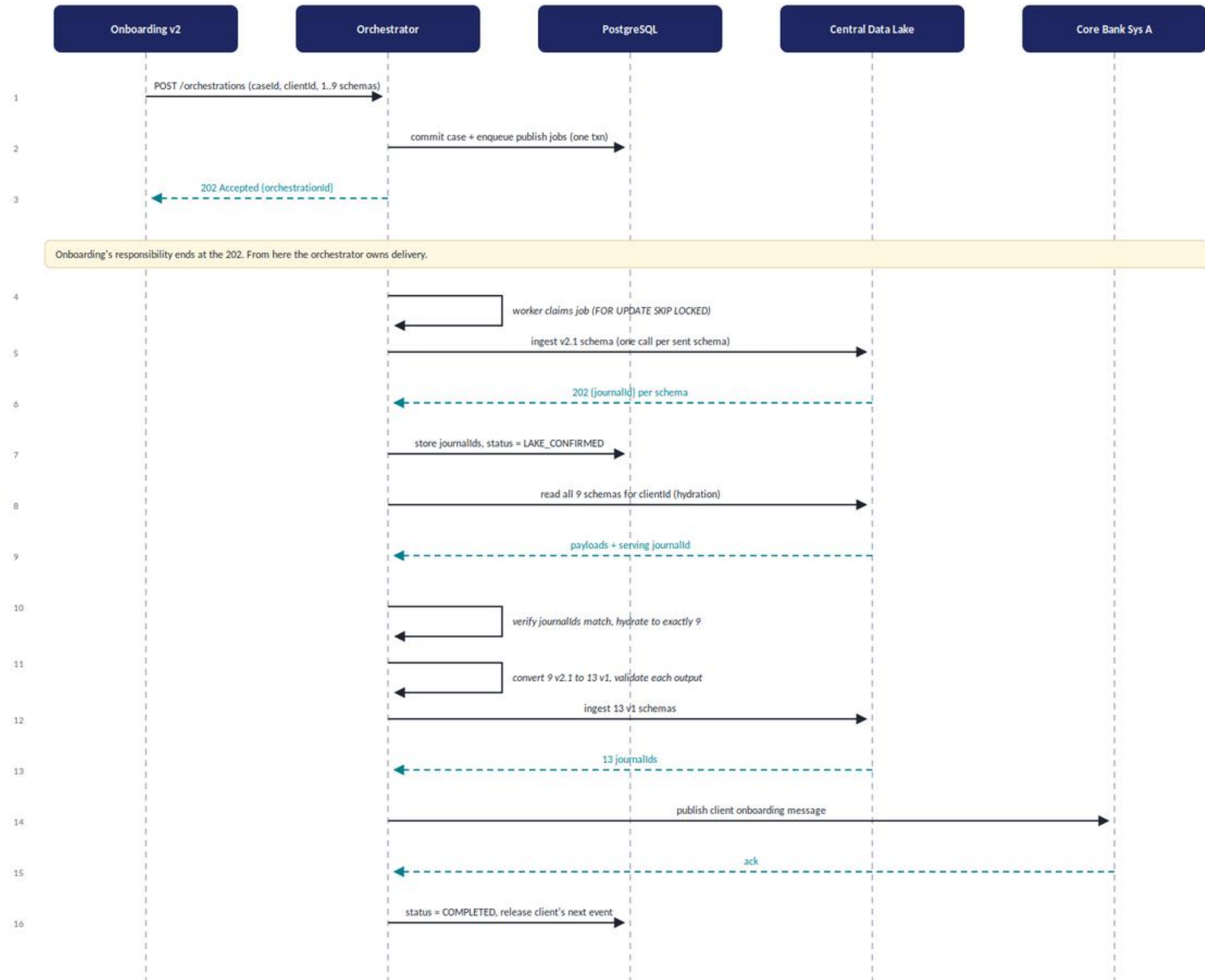


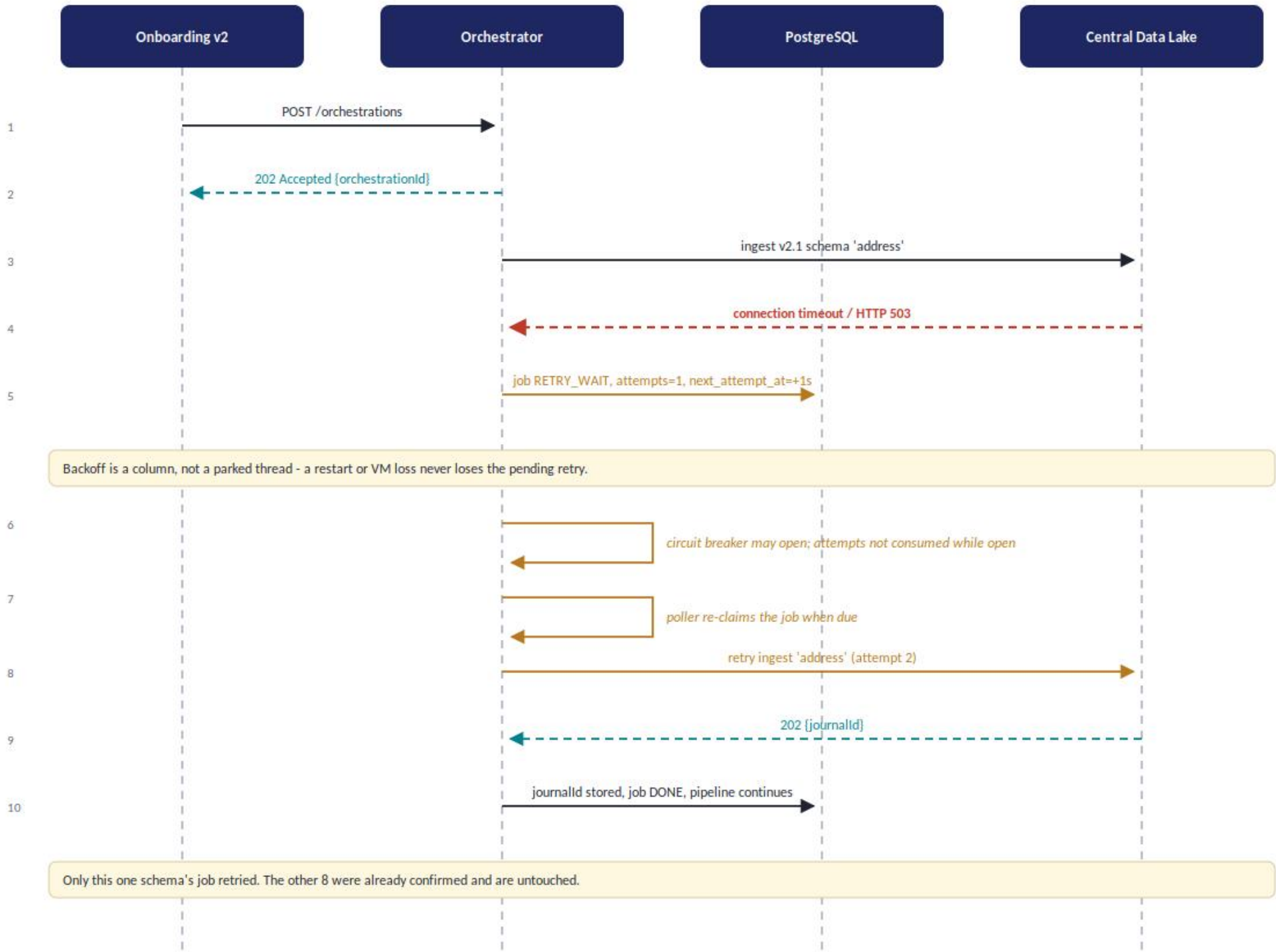
Scenario 1 - Happy Path (end to end)

9 schemas in, 13 out, CBS-A in parallel, client's next event released



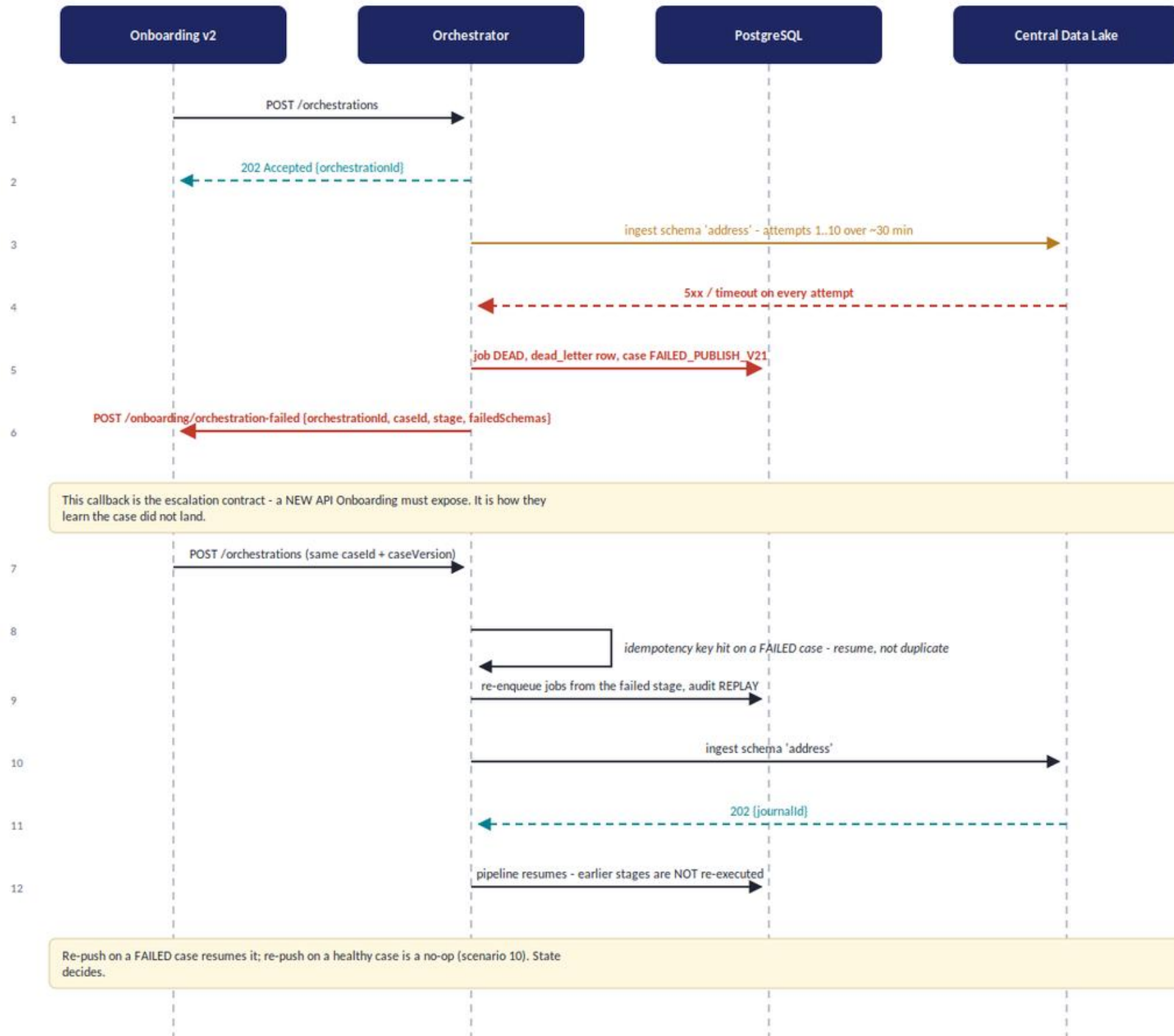
Scenario 2 - Data Lake Push Fails (network / 5xx, transient)

Self-heals inside the orchestrator. Onboarding is never told, and no re-push is needed.



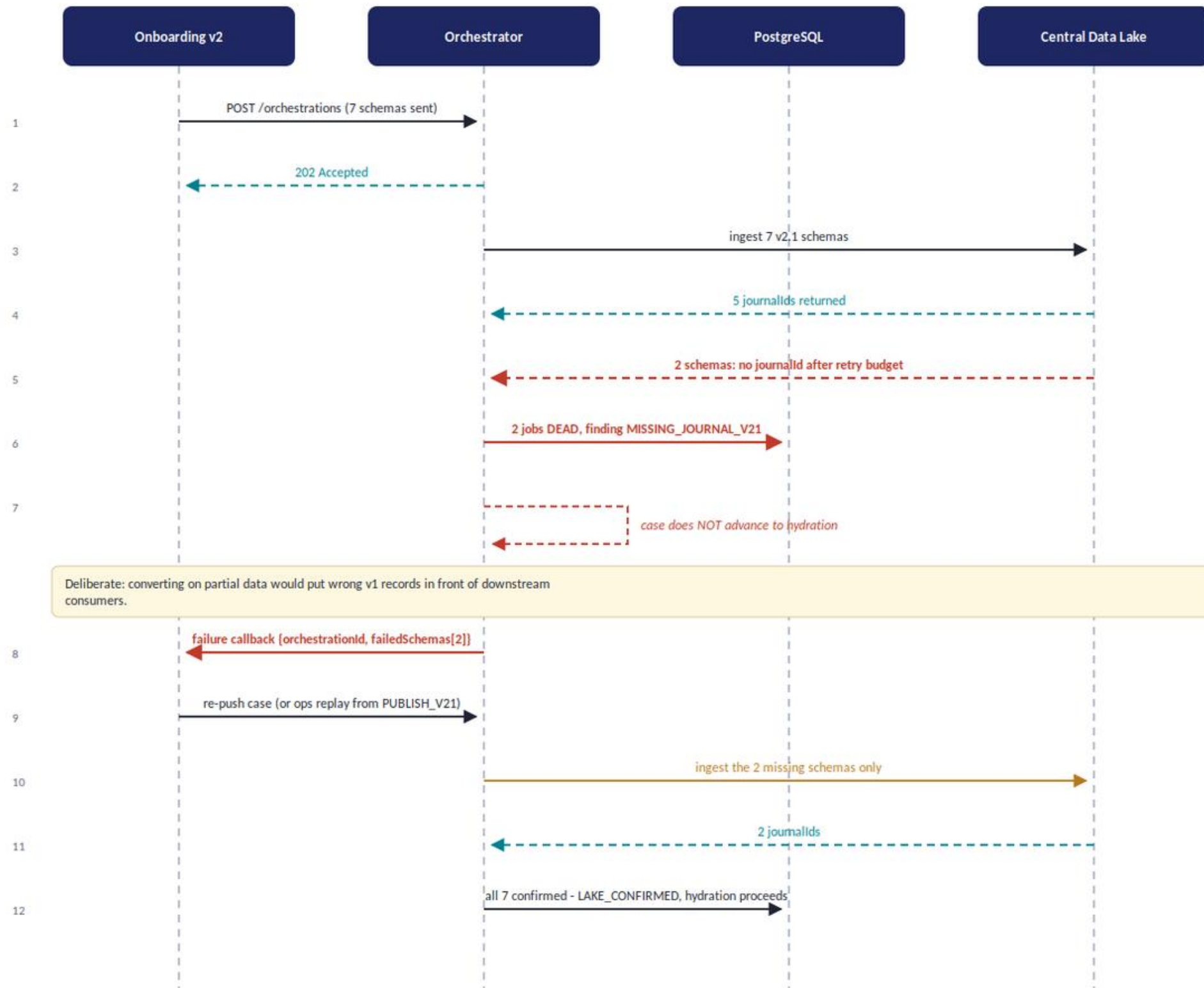
Scenario 3 - Lake Push Fails Permanently, Onboarding Asked to Re-push

Retries exhausted: dead-letter, then escalate the orchestrationId back to Onboarding



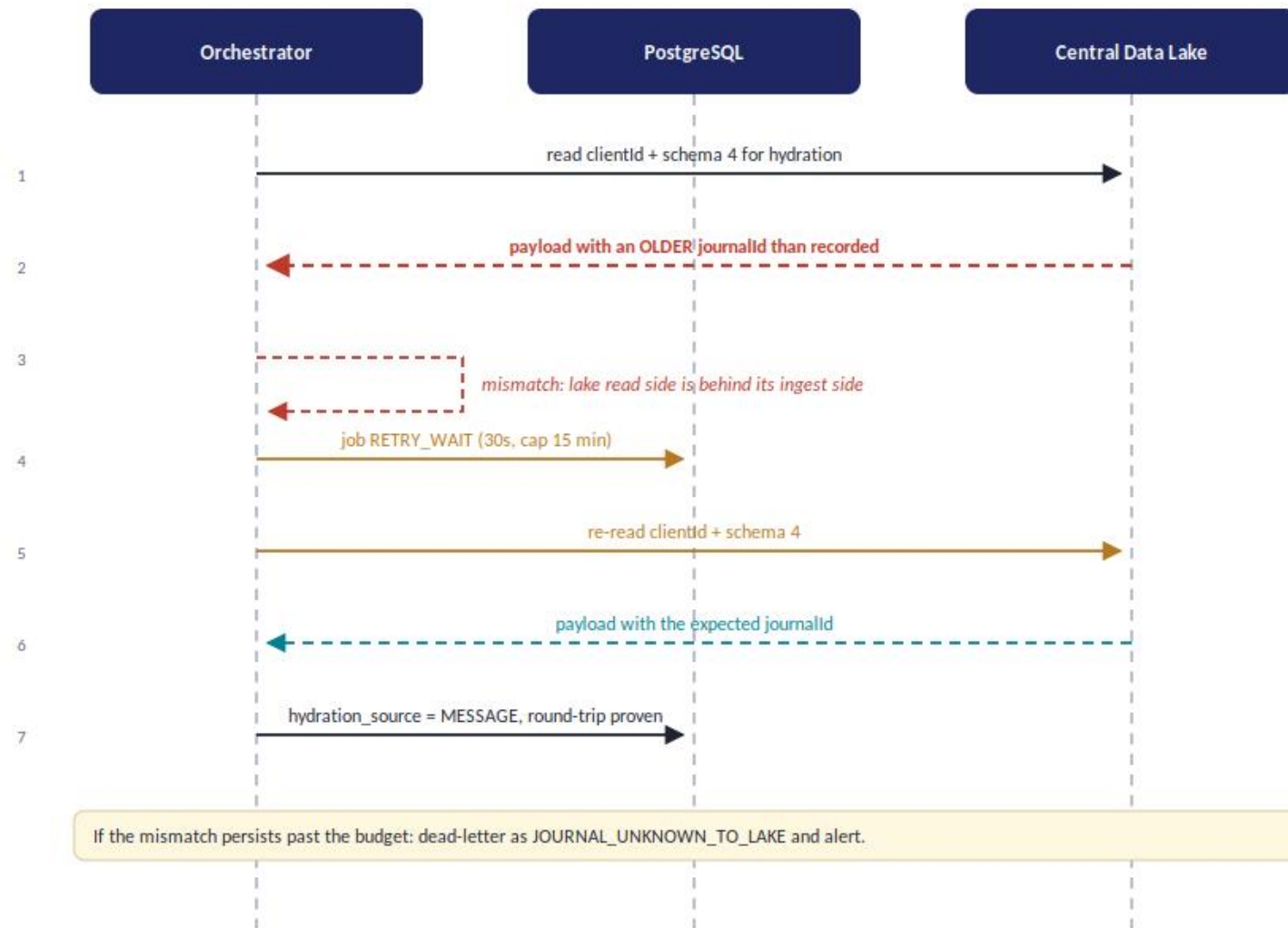
Scenario 4 - Only Some Schemas Confirmed by the Lake

Blocked before conversion: no journalId, no progress



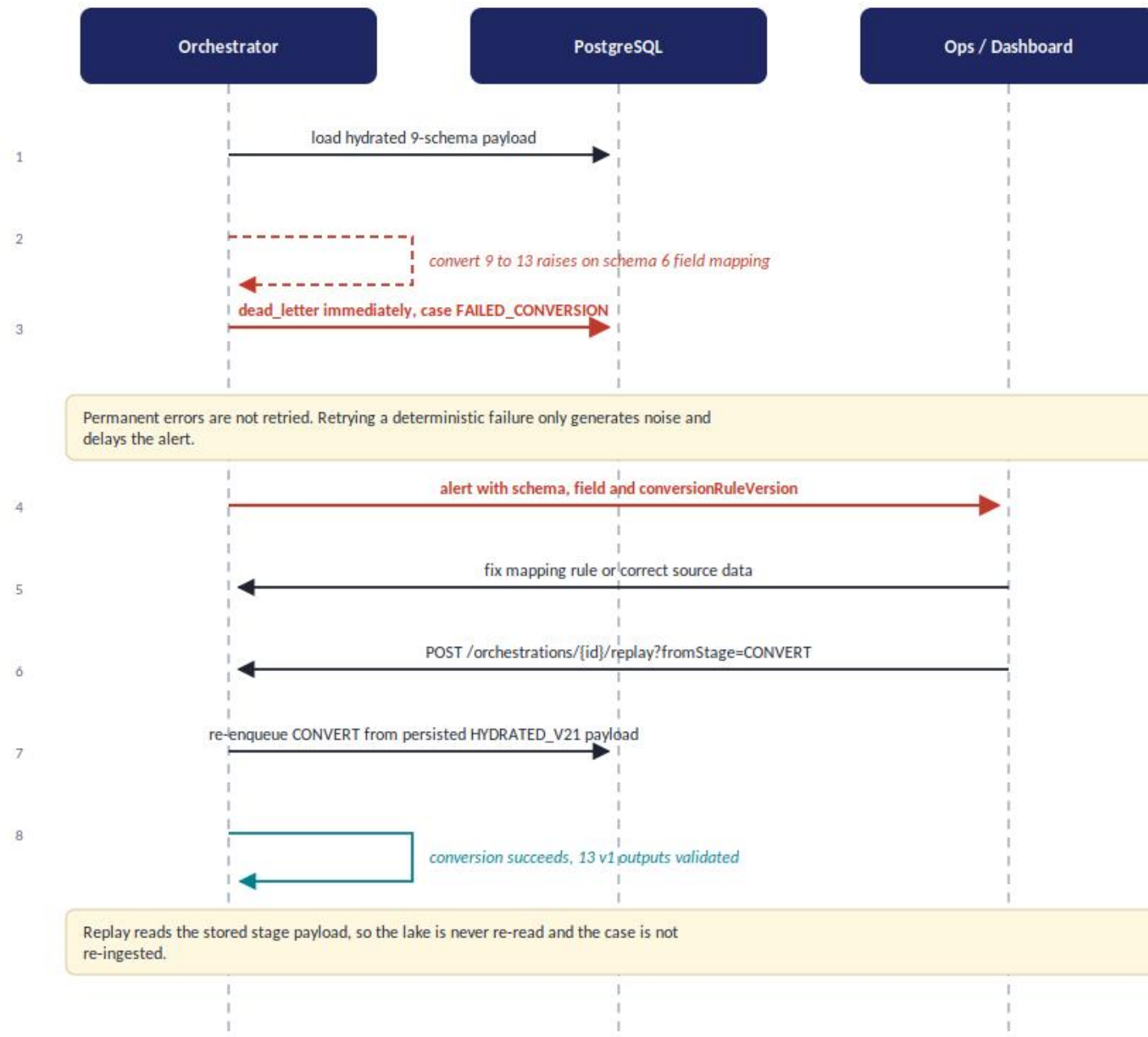
Scenario 5 - Lake Read Lags Behind Ingest (journalId mismatch)

Self-heals: the hydrator refuses to convert data the lake has not made visible yet



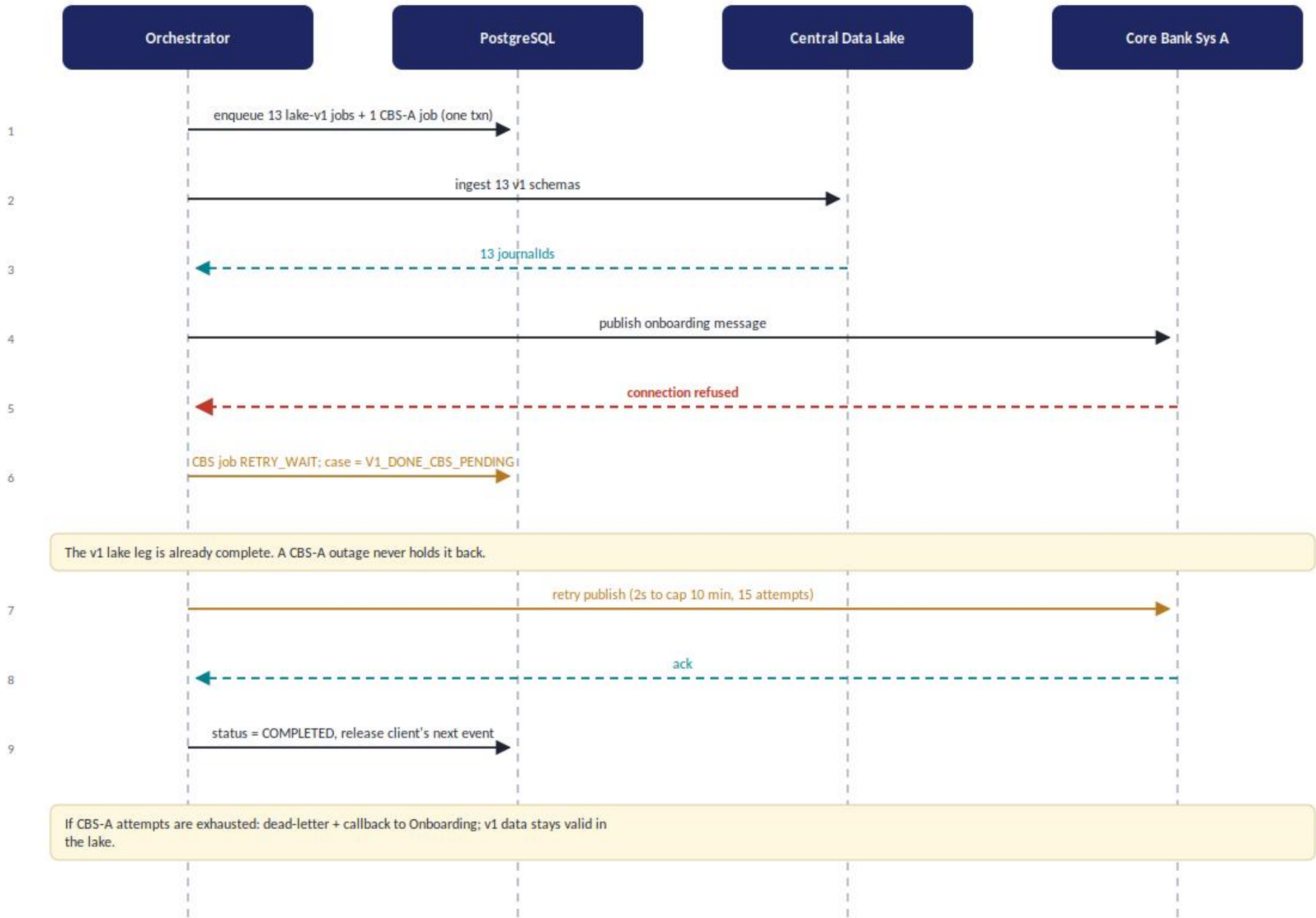
Scenario 6 - Conversion Fails (permanent data / rule error)

No retry: a deterministic failure will fail identically every time



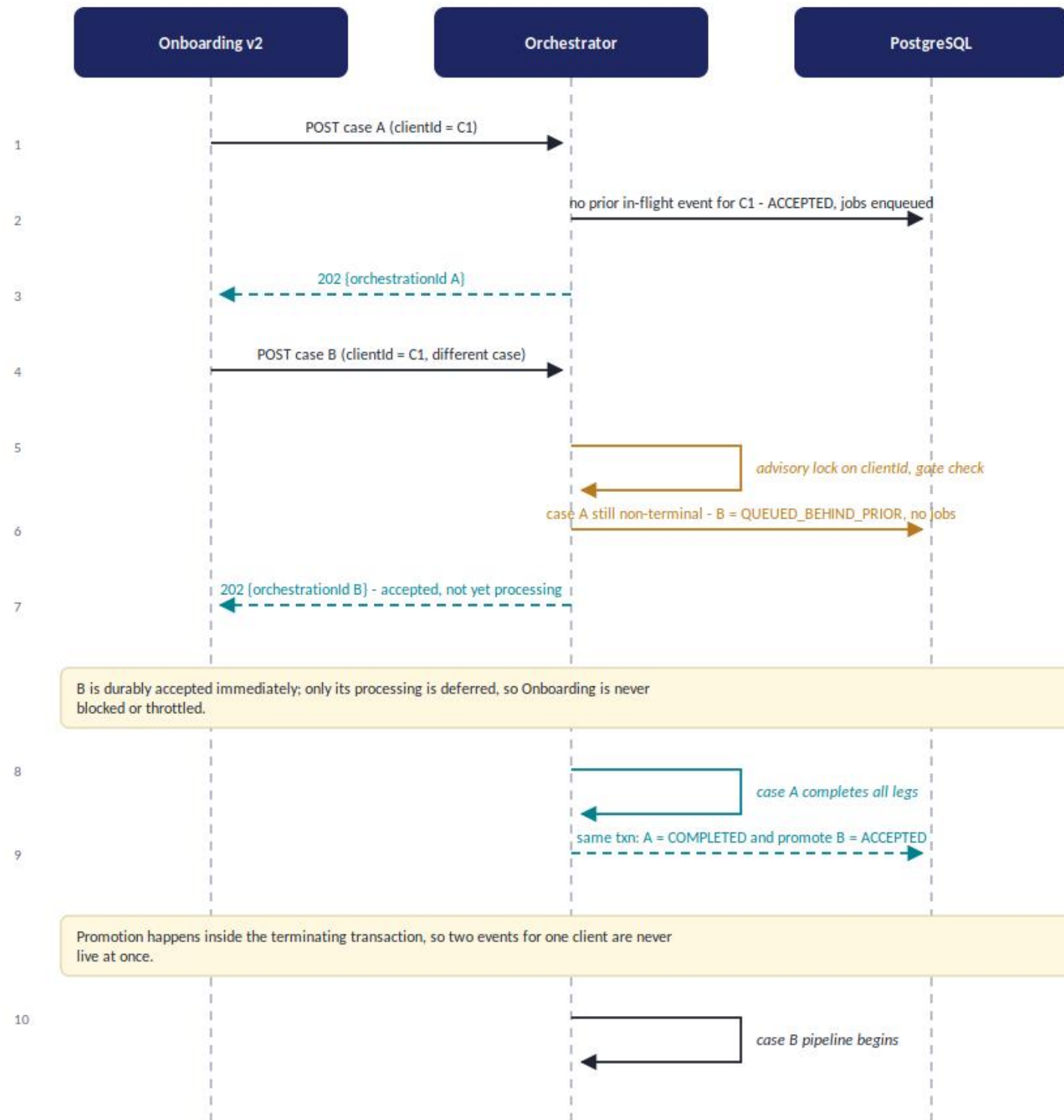
Scenario 7 - CBS-A Down While the v1 Lake Leg Succeeds

The two final legs are independent jobs with independent retry budgets



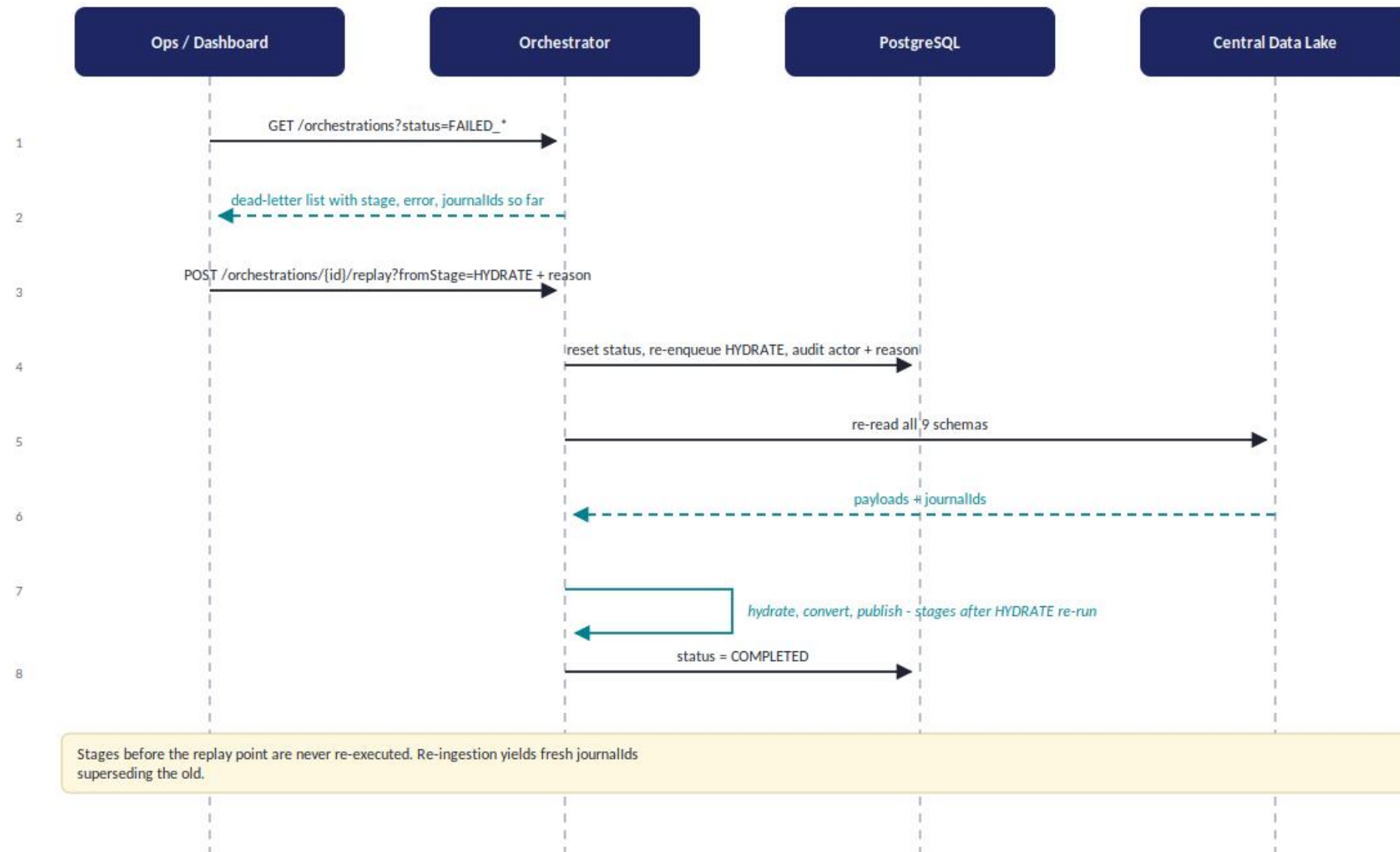
Scenario 8 - Second Case for the Same Client Arrives Mid-flight

Per client, event N must fully complete before N+1 starts - across cases



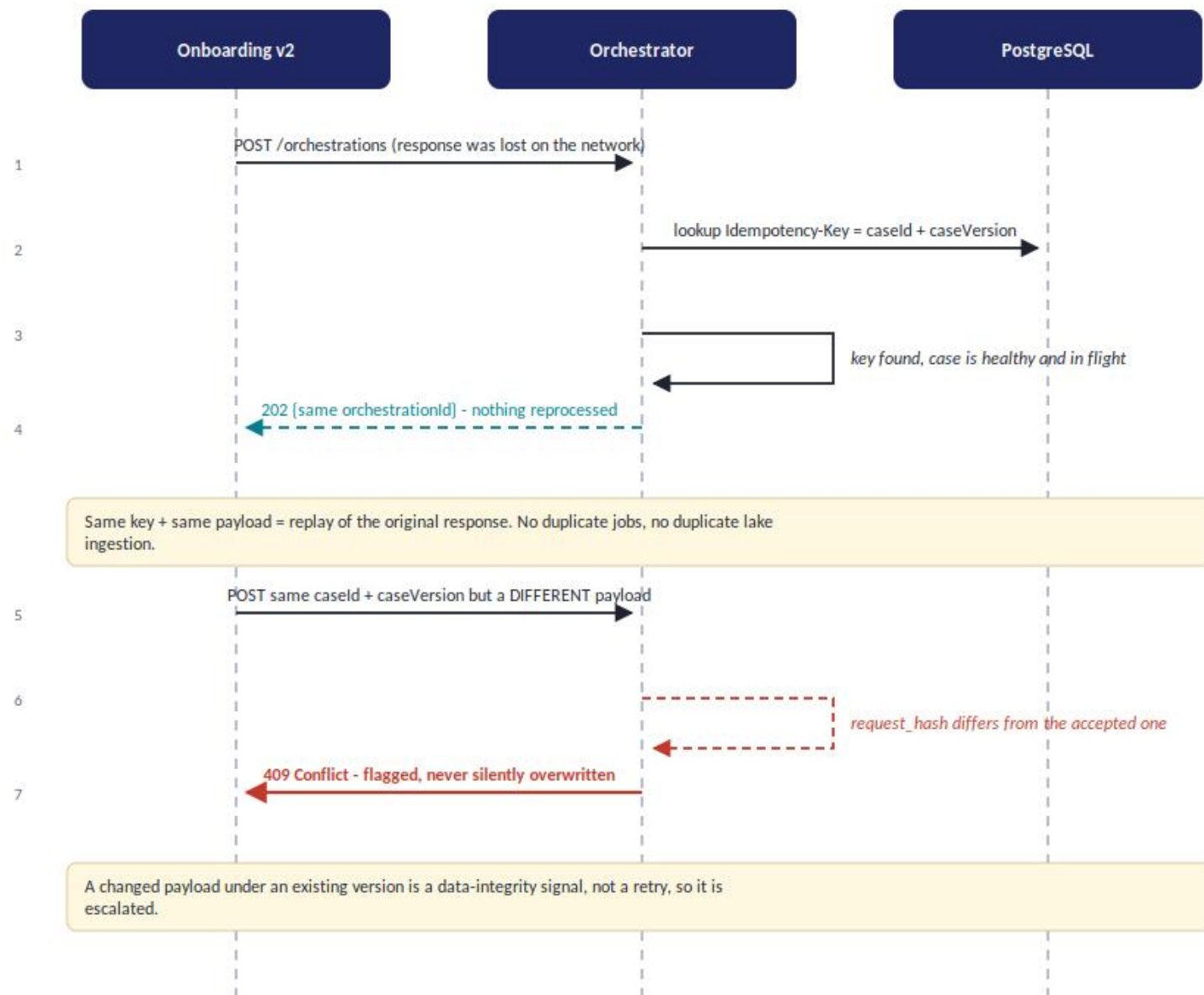
Scenario 9 - Ops Replay From the Dashboard

Every stage is idempotent and re-runnable from its persisted input



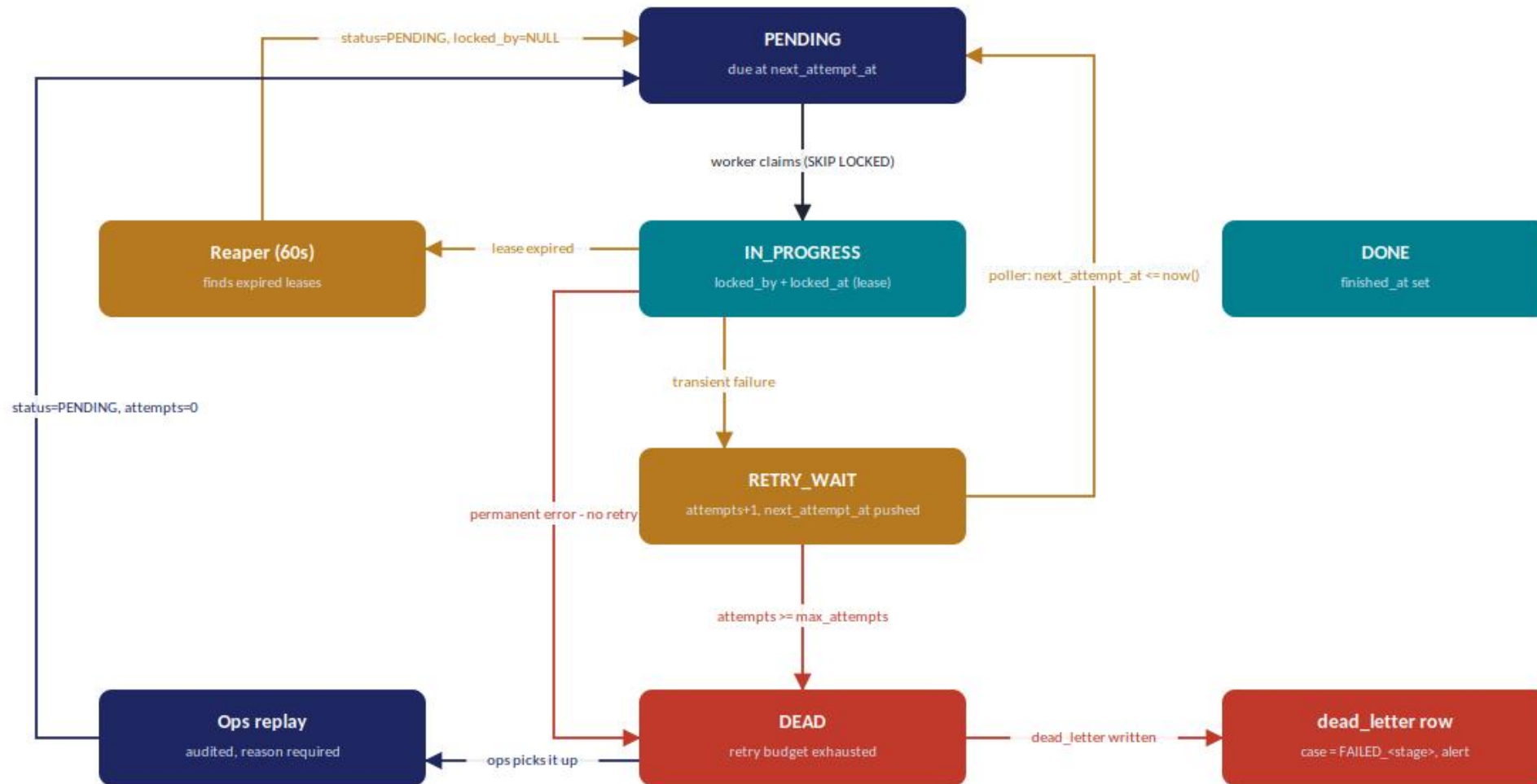
Scenario 10 - Onboarding Re-pushes a Case That Is Already Healthy

Idempotency: a caller retry after a lost response must never create a second pipeline



Job Row Lifecycle - state and column changes

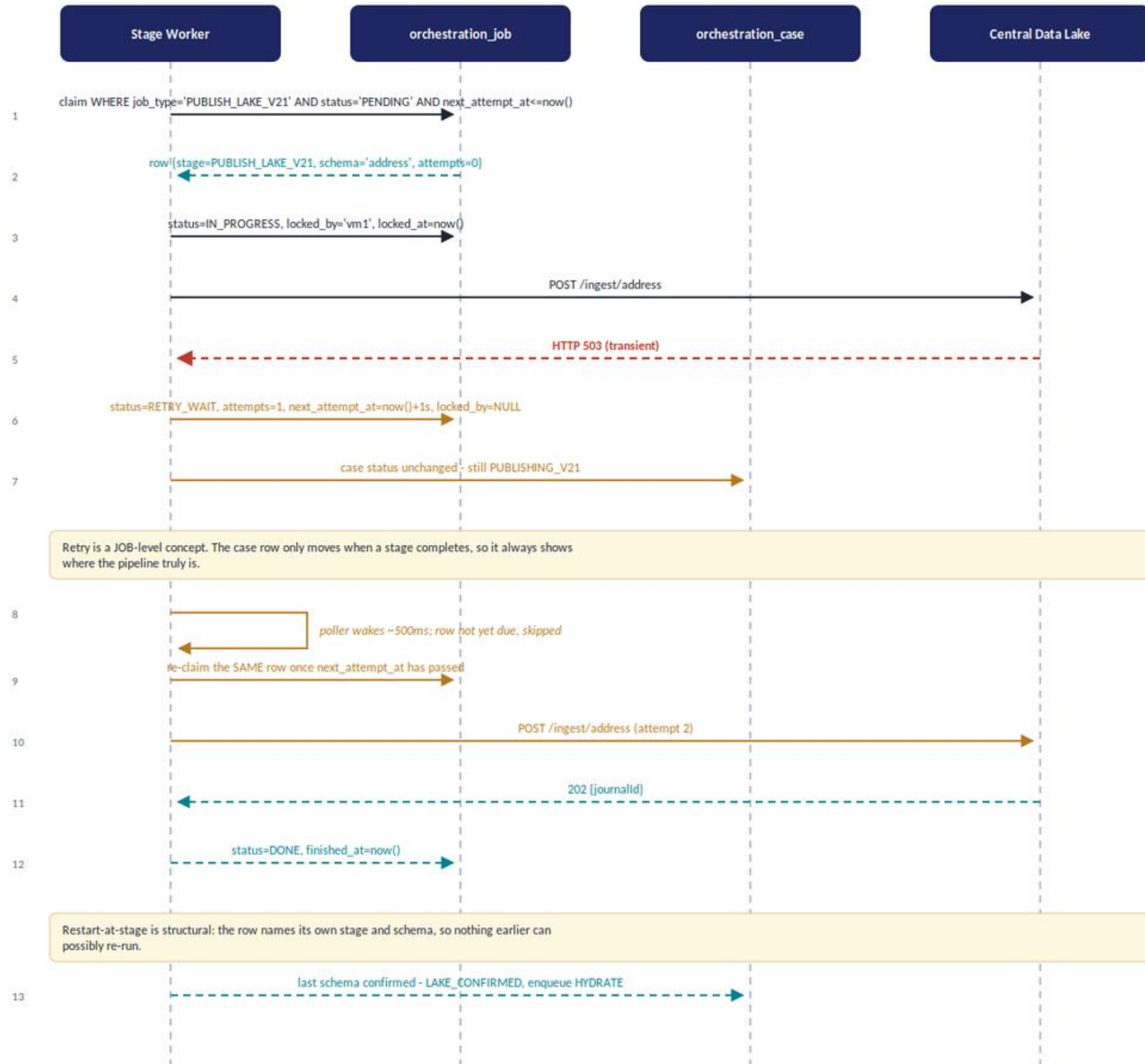
One row in orchestration_job. Retry, orphan recovery and replay are all transitions on this row.



Retry, orphan recovery and replay all converge on PENDING - one execution path to test and monitor. Retry = two column writes, so a restart never loses a pending attempt. Orphan recovery = lease expiry, so a dead VM cannot strand work (attempts is NOT incremented - a crash is not a failed attempt). Replay = the same reset, triggered by a human instead of a timeout.

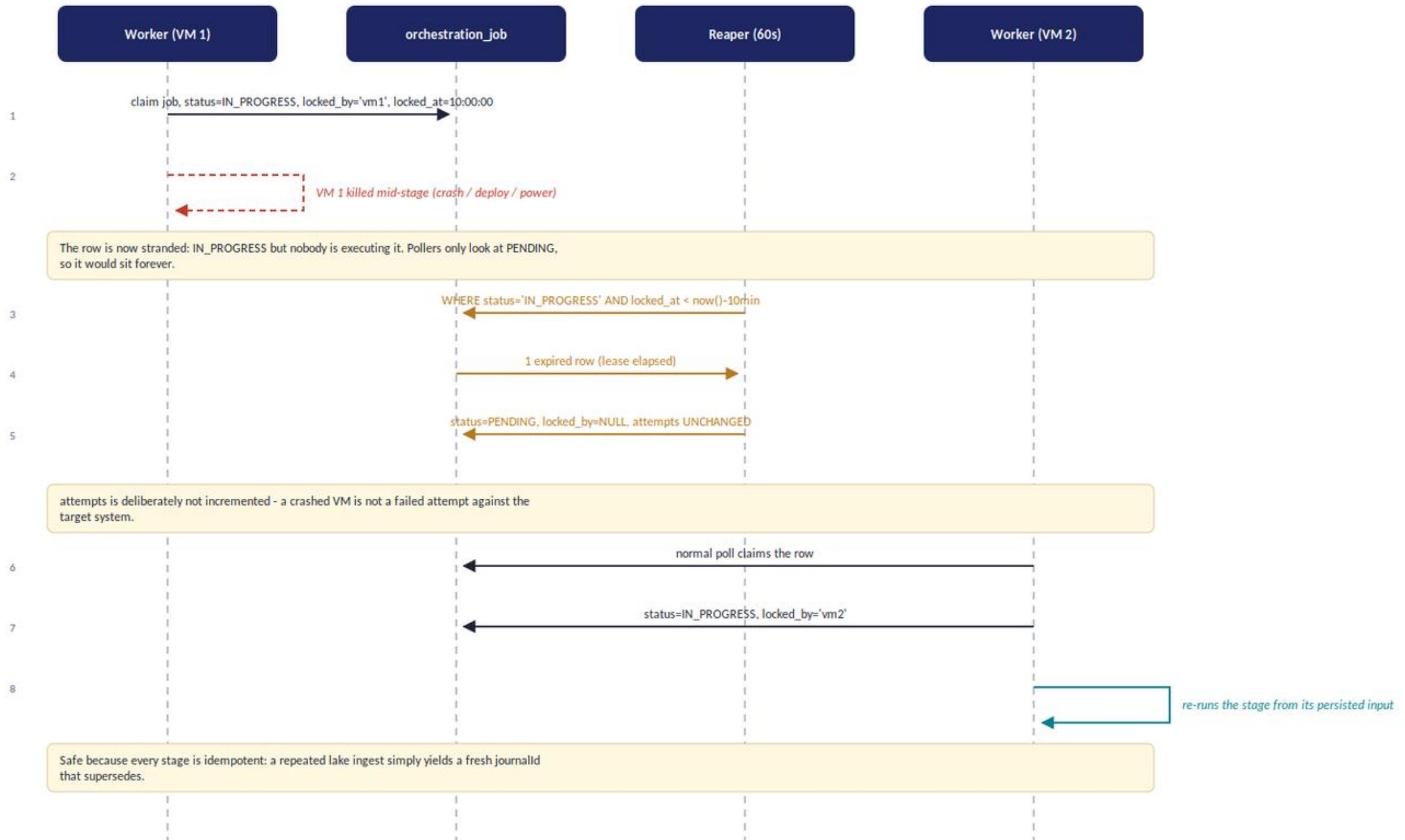
Retry - How a Failed Stage Restarts at That Same Stage

The job row carries the stage identity, so a retry can only re-run the stage that failed



Retriever - Recovering a Job Whose Worker Died Mid-stage

Lease expiry is what makes a crash recoverable without human involvement



Replay - Restarting a Dead Case From a Chosen Stage

Replay reuses the persisted stage input, so stages before the replay point never re-run

